Empirical Proof Record: VALIDATED

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Proof ID: `4a08ba3f5f8d9d9aade8466bf9df88f9…`
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1. Claim

> Across a trajectory with spaced re-exposures, the substrate keeps recognising every re-exposed stimulus: (for every same-stimulus re-exposure pair (i, j) in the trajectory, Jaccard(concepts_i, concepts_j) $\geq$ 0.80). The reported floor is the worst pair across the whole run.

- **Operationalisation:** Run a schedule of len(stimuli) unique stimuli each repeated 3× and interleaved (gap = len(stimuli)) through Kimera's entity target (Takwin). For each stimulus, take the per-cycle "concepts" set at each exposure and compute Jaccard over all exposure pairs. recognition_jaccard_floor = min over all pairs across all stimuli; the LTL invariant holds iff floor $\geq \theta$.
- **Threshold:** `recognition_jaccard_floor >= 0.8 jaccard`
- **H0:** H0: floor < 0.80 — recognition collapses under manifold deformation somewhere in the run
- **H1:** H1: floor $\geq$ 0.80 — recognition is stable across the whole trajectory (memory-as-deformation holds as a flow property)

2. Verdict

VALIDATED — observed `0.9473684210526315`
recognition floor 0.9474 over 21 re-exposure pairs (8 stimuli × 3 exposures); mean Jaccard 0.9950; 3 exposures produced no concept set; worst pair: stimulus 0 cycles 0↔8 = 0.9474

3. Evidence

Pillar	Statistic	Value	95% CI	Lib
`O.flow.recognition_stability`	`recognition_jaccard_floor`	`0.9473684210526315`	—	oph

4. Pre-registration

- Registered at: `2026-05-21T23:01:12.607321+00:00`
- Config hash: `0fbe8ce56b2a2db7203910b72aaebc0c…`
- Data hash: `f36c007933b4b5223d517967d58c7b80…`

5. Signature

`70fc8834d019e2adae6915f6550ce637e14f7b21549d8e5b6b0e371e0551f83d`